



HUMAN RESOURCE PLAN TEMPLATE

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HUMAN RESOURCE PLAN DOST STARBOOKS: WHIZ CHALLENGE

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INTRODUCTION

This Human Resource Management Plan serves as a guiding document for managing all human resource activities from project initiation through closure. It defines how the team is structured, how responsibilities are assigned, and how personnel will be supported throughout the project lifecycle.

The STARBOOKS Whiz Challenge is a gamified, offline, quiz-based educational platform developed in collaboration with DOST-STII. The project involves a multidisciplinary development team responsible for requirements analysis, UI/UX design, backend and frontend development, quality assurance, and stakeholder liaison. Given the project's scope — which includes 4 game modes (Whiz Challenge, Memory Match, Jigsaw Puzzle, and 1v1 Battle), player profiling, an admin dashboard, and gamification mechanics — clear human resource management is essential to ensure coordinated effort and timely delivery.

This Human Resource Management Plan includes:

- Roles and responsibilities of all project team members
- Project organizational structure and RACI Matrix
- Staffing management plan covering:
 - a. How resources will be acquired
 - b. Timeline for resources/skill sets aligned with project phases
 - c. Training required to address skill gaps
 - d. How performance reviews will be conducted
 - e. Recognition and rewards system

The purpose of this plan is to ensure the right people are assigned to the right tasks, that communication and accountability are clearly established, and that the team is empowered to deliver a high-quality product that meets the expectations of DOST-STII.

ROLES AND RESPONSIBILITIES

Roles and responsibilities must be clearly defined to ensure every team member understands their scope of authority, accountability, and contribution. The STARBOOKS Whiz Challenge project team consists of six defined roles. Each role description includes the member's authority level, primary responsibilities, and required competencies.



Project Manager (PM) — 1 Position

The Project Manager is accountable for the overall success of the STARBOOKS Whiz Challenge project. The PM holds the highest level of authority within the project team and is responsible for approving deliverables, managing the project schedule and scope, and ensuring alignment with DOST-STII's expectations.

Authority: The PM has final approval authority over project deliverables, scope decisions, and resource assignments. The PM is the primary point of escalation for unresolved issues within the team.

Responsibilities:

- Define and manage project scope, objectives, and milestones
- Lead requirements gathering efforts and coordinate with the DOST-STII liaison
- Oversee the selection and finalization of the technology stack in coordination with the Backend and Frontend Developers
- Authorize and approve deployment builds prior to release
- Monitor team performance and ensure adherence to the project schedule
- Serve as the primary decision-maker for the project and escalation point for risks or blockers
- Report project status regularly and facilitate team meetings and retrospectives

Competencies: Leadership and team management, project scheduling, scope management, stakeholder communication, risk management, and familiarity with software development processes.

UI/UX Designer — 1 Position

The UI/UX Designer is responsible for the visual design and user experience of the STARBOOKS Whiz Challenge application. This role ensures the product is intuitive, visually engaging, and appropriate for the target audience — primarily elementary to senior high school students interacting with the kiosk system at DOST events and exhibits.

Authority: The UI/UX Designer has decision-making authority over visual design and user experience direction within the agreed design guidelines, subject to review and approval by the Project Manager.



Responsibilities:

- Design wireframes, prototypes, and high-fidelity mockups for all application screens, including game modes, player stats, and the admin dashboard
- Review and finalize the system design in coordination with backend and frontend developers
- Lead mockup testing sessions with stakeholders, including the DOST-STII team
- Incorporate feedback from DOST-STII into revised design iterations
- Consult on frontend development to ensure visual fidelity is maintained during implementation
- Maintain design assets and ensure consistency in typography, iconography, color system, and branding across all screens

Competencies: Proficiency in Figma or equivalent prototyping tools, UI/UX principles, user-centered design methodologies (including Design Thinking), visual communication, and familiarity with game interface design for young audiences.

Frontend Developer — 1 Position

The Frontend Developer is responsible for implementing the user-facing interface of the STARBOOKS Whiz Challenge application, translating the UI/UX mockups into a functional, responsive, and performant experience.

Authority: The Frontend Developer has authority over frontend implementation decisions. Changes to interface design or behavior must be validated against the approved UI/UX mockups and coordinated with the UI/UX Designer and Backend Developer.

Responsibilities:

- Implement all player-side functionalities, including game modes (Whiz Challenge, Memory Match, Jigsaw Puzzle, 1v1 Battle), badge system, player profiles, and statistics screens
- Develop the admin dashboard frontend in coordination with the Backend Developer
- Integrate gamification features such as the level map, badge rewards, difficulty progression, and answer explanations
- Perform API integration with backend services
- Collaborate with the UI/UX Designer to faithfully translate mockup designs into code
- Participate in QA and testing phases to address UI bugs and polish the experience post-feedback
- Assist in preparing the deployment build alongside the Backend Developer



Competencies: Frontend development (e.g., React, Vue, or equivalent framework), HTML/CSS, JavaScript, responsive and component-based design, API integration, and cross-platform compatibility for kiosk environments.

Backend Developer — 1 Position

The Backend Developer is responsible for the server-side architecture, database design, API development, and AI feature integration for the STARBOOKS Whiz Challenge platform. This role ensures the application is stable, secure, and performant — particularly in offline environments.

Authority: The Backend Developer is accountable for all backend architecture decisions and has authority over database design and setup. Technology stack choices require coordination and approval from the Project Manager.

Responsibilities:

- Design and implement the backend architecture for all game modes and the admin dashboard
- Design and set up the database structure for player accounts, scores, badges, question banks, and analytics
- Develop and maintain REST APIs for communication between the frontend and backend
- Integrate the AI-assisted quiz question generation feature
- Prepare the deployment build in coordination with the Frontend Developer
- Execute deployment and installation of the application on STARBOOKS kiosk setups
- Resolve bugs and performance issues identified during QA testing and user testing
- Consult the Project Manager and UI/UX Designer on technical feasibility during planning phases

Competencies: Backend programming (e.g., Node.js, Python, or equivalent), database management (SQL/NoSQL), REST API development, system architecture, offline-first application design, and AI/ML integration fundamentals.

QA/Tester — 1 Position

The QA/Tester is responsible for ensuring the quality, reliability, and usability of the STARBOOKS Whiz Challenge application prior to deployment. This role validates that all features function as intended, meet defined requirements, and deliver a smooth user experience for both players and administrators.



Authority: The QA/Tester has authority to identify defects, flag non-conforming deliverables, and require fixes before sign-off on any module. Final approval to advance from testing phases rests with the Project Manager.

Responsibilities:

- Conduct functional testing across all game modes, difficulty levels, scoring systems, badge logic, and admin dashboard features
- Coordinate and facilitate user testing sessions with DOST-STII representatives and target users (students, administrators)
- Document test results, defect reports, and performance observations
- Work with the Backend and Frontend Developers to verify bug fixes and validate resolved issues
- Ensure all features meet the acceptance criteria established in the requirements phase before deployment

Competencies: Software testing methodologies (functional, regression, usability), defect tracking and reporting, test case design, user acceptance testing (UAT) facilitation, and attention to detail.

Liaison to DOST-STII — 1 Position

The Liaison to DOST-STII serves as the primary communication bridge between the development team and the client organization. This role ensures that DOST-STII's requirements, expectations, and feedback are accurately conveyed to the team and that deliverables are properly presented and accepted.

Authority: The Liaison acts as the accountable party for all client-facing communications and the final demo and documentation submission. This role does not hold technical authority but has significant influence over requirements and acceptance criteria.

Responsibilities:

- Gather and document requirements from DOST-STII stakeholders (Ms. Marievic V. Narquita and Mr. McGyver C. Basaya)
- Facilitate and coordinate all meetings, site visits, and feedback sessions with DOST-STII
- Communicate client requirements, constraints, and feedback to the Project Manager and team
- Represent the project team during user testing with DOST representatives and target users
- Oversee the preparation of the final demo presentation and project documentation for submission to DOST-STII



- Ensure all deliverables align with DOST-STII's mission of accessible, innovative STEM education

Competencies: Stakeholder management, professional communication, documentation, requirements elicitation, and knowledge of DOST-STII's STARBOOKS program and objectives.

PROJECT ORGANIZATIONAL CHARTS

The STARBOOKS Whiz Challenge project follows an Agile collaborative team structure where the Project Manager holds overarching accountability while each specialist role operates with functional autonomy within their defined domain. The DOST-STII Liaison bridges the team with the client organization. The RACI Matrix below defines task-level responsibilities for all major project deliverables, drawn directly from the project's documented activity list:

	Project Manager	UI/UX Designer	Backend Developer	Frontend Developer	QA/Tester	Liaison to DOST-STII
Define project scope and objectives	A	C	C	C	I	R
Requirements gathering	R	C	C	C	I	A
Create use case diagrams & documentation	R	C	C	C	I	C
Design wireframes and mockups	I	R	C	C	I	C
Review and finalize system design	A	R	C	C	C	C
Select and finalize tech stack	R	C	A	A	I	C
Backend development	I	I	R	C	I	I
Frontend development	I	C	C	R	I	I
API integration	I	I	R	R	I	I
Database design and setup	I	I	A	C	I	I
Implement gamification features	I	C	C	R	C	I



Create admin dashboard	I	C	R	R	I	C
Develop player-side functionalities	I	C	R	R	I	I
Conduct functional testing	I	I	C	C	R	I
Conduct user testing (w/ DOST and target users)	I	I	I	I	R	A
Fix bugs and polish based on feedback	I	C	R	R	R	I
Prepare deployment build	R	I	A	A	C	I
Deployment and installation	A	I	R	R	I	C
Final demo to DOST and documentation submission	A	I	C	C	C	R

Key:

R – Responsible for completing the work

A – Accountable for ensuring task completion/sign off

C – Consulted before any decisions are made

I – Informed of when an action/decision has been made

STAFFING MANAGEMENT

Staff Acquisition:

The Project Manager is responsible for coordinating role assignments and ensuring that each team member's skills are appropriately matched to their designated function. All six roles — Project Manager, UI/UX Designer, Backend Developer, Frontend Developer, QA/Tester, and Liaison to DOST-STII — were assigned at project kickoff.

The team is not co-located full-time; coordination is conducted primarily through online collaboration tools (e.g., Canva Whiteboards, Figma, and group messaging platforms), with scheduled onsite meetings at DOST-STII as required by project milestones.



Resource Calendars:

The project spans approximately fourteen months, from April 2025 through June 2026, and is divided into four major phases as defined in the project roadmap. Resource involvement and workload intensity are mapped across each phase as follows:

Phase 1 — Requirements Gathering and Planning (April – August 2025)

This phase focuses on client engagement, scope definition, requirements documentation, system diagramming, and initial wireframe design. The Liaison to DOST-STII and Project Manager carry the heaviest workloads during this period, facilitating stakeholder interviews with the STARBOOKS team (Ms. Marievic V. Narquita and Mr. McGyver C. Basaya), defining core features, and drafting the project timeline. The UI/UX Designer is engaged in producing initial wireframes. Backend and Frontend Developers participate in an advisory capacity to inform technical feasibility, while the QA/Tester has minimal involvement at this stage.

Role	Workload Intensity
Project Manager	High
Liaison to DOST-STII	High
UI/UX Designer	Moderate
Backend Developer	Low
Frontend Developer	Low
QA/Tester	Low

Phase 2 — Development Phase 1 (September – October 2025)

This phase covers backend and frontend implementation of foundational system components: finalizing the system architecture, setting up the development environment, building the registration and login system, developing the homepage and player profile, and creating the Memory Match and Puzzle game modes. The backend and frontend developers are at peak workload. The UI/UX Designer remains actively engaged to ensure implementation fidelity to approved mockups. The Project Manager monitors progress and coordinates deliverables. The QA/Tester begins preliminary functional reviews of completed modules.

Role	Workload Intensity
Backend Developer	High
Frontend Developer	High
UI/UX Designer	Moderate
Project Manager	Moderate
QA/Tester	Low–Moderate



Liaison to DOST-STII	Low
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Phase 3 — Development Phase 2 (November 2025 – March 2026)

This is the most intensive development phase, encompassing the Whiz Challenge and 1v1 Battle game modes, the leaderboard system, the admin dashboard, analytics and leaderboard tracking, and internal testing. The Backend and Frontend Developers remain at their highest sustained workload across this phase. The QA/Tester's involvement escalates significantly as internal testing begins. The Project Manager actively oversees multiple parallel workstreams and manages scope and schedule risks. The UI/UX Designer provides continued support for interface refinements.

Role	Workload Intensity
Backend Developer	High
Frontend Developer	High
QA/Tester	High
Project Manager	High
UI/UX Designer	Moderate
Liaison to DOST-STII	Low

Phase 4 — Testing, Deployment, and Finalization (April – June 2026)

This final phase encompasses system-wide testing, user acceptance testing (UAT) with DOST-STII and target users, bug fixing, UI/UX polish, final deployment, and submission of all project deliverables. The QA/Tester leads this phase operationally, coordinating closely with the Backend and Frontend Developers for defect resolution. The Liaison to DOST-STII returns to a high-engagement level to facilitate UAT sessions and coordinate the final demo and documentation submission. The UI/UX Designer handles final interface polish. The Project Manager oversees project closure activities.

Role	Workload Intensity
QA/Tester	High
Liaison to DOST-STII	High
Backend Developer	High
Frontend Developer	High
Project Manager	High
UI/UX Designer	Moderate



Training:

At project initiation, the team conducted a collective assessment of skills relevant to the project's technical and operational requirements. The following training needs were identified:

- **Offline-first application architecture** — Given that the STARBOOKS Whiz Challenge must run entirely offline on STARBOOKS kiosk hardware, the Backend and Frontend Developers undertook self-directed learning on offline-first design patterns, local data persistence, and environment setup for kiosk deployment — addressed ahead of Development Phase 1.
- **Gamification mechanics design** — The team supplemented their knowledge through reference materials on badge systems, reward loops, leaderboard design, and engagement strategies appropriate for the student audience, in preparation for Development Phase 2.
- **Figma proficiency** — Members involved in wireframing and mockup development ensured fluency in Figma during the Requirements Gathering and Planning phase to support collaborative prototype design.
- **Analytics and dashboard development** — The Backend and Frontend Developers familiarized themselves with data aggregation and visualization techniques in preparation for the admin dashboard and analytics tracking features in Development Phase 2.

Performance Reviews

The Project Manager is responsible for monitoring and evaluating the performance of all team members throughout the project lifecycle. Performance evaluation is based on the following criteria:

- **Timeliness** — Whether assigned tasks and deliverables are completed within agreed phase timelines
- **Quality of output** — Whether work products meet defined acceptance criteria and technical or design standards
- **Collaboration** — The effectiveness of communication and cooperation with team members and DOST-STII stakeholders
- **Initiative and problem-solving** — The degree to which team members proactively identify and resolve issues within their domain
- **RACI compliance** — Whether team members are fulfilling their designated Responsible, Accountable, Consulted, and Informed duties as defined in the RACI Matrix



At the conclusion of each major roadmap phase, the Project Manager will conduct a brief informal review to assess team progress and provide constructive feedback. A comprehensive performance review will be conducted at project closure (June 2026), at which time the Project Manager will document each member's contributions and communicate results to the course instructor and project adviser.

SPONSOR ACCEPTANCE

Approved by the Project Sponsor:

Marievic V. Narquita
Information Services Section (ISS) Head of DOST-STII

Date: April 8, 2026

